



A STUDY OF ADOPTION INTENTION OF ONLINE-PHARMACIES AMONG COIMBATORE
RESIDENTS

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21MBA143

Under the guidance of

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A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled **“A STUDY OF ADOPTION INTENTION OF ONLINE-PHARMACIES AMONG COIMBATORE RESIDENTS”** is for course completion of Research Project I is the Bonafide work of **SNEHA S** who carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION

I hereby declare that this Research project report entitled as, “**A STUDY OF ADOPTION INTENTION OF ONLINE-PHARMACIES AMONG COIMBATORE RESIDENT**” has been undertaken for academic purpose for the course submitted to Anna University in partial fulfilment of requirement for the award of degree of Master of Business Administration. The project report is the record of the original work done by me under the guidance of Mr. VIVEK RAJ S N, Assistant Professor, KCT Business School during the academic year 2021-2023.

I, also declare hereby, that the information given in this report is correct to the best of my Knowledge and behalf.

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Name and Signature

Date:

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LIST OF TABLES

TABLE NUMBER	TITLE	PAGE NUMBER
1	MULTIPLE TEST OF ANN	41
2	ACCURACY OF ANN	42
3	INDEPENDENT VARIABLE IMPORTANCE OF ANN	43
4	ACCURACY OF LOGISTIC REGRESSION FOR ADOPTION INTENTION OF ONLINE PHARMACIES	47
5	LOGISTIC REGRESSION FOR INTENTION TO ADOPT ONLINE PHARMACIES	48
6	ACCURACY OF DECISION TREE	50
7	MODEL SUMMARY RESULT OF DECISION TREE	50

LIST OF FIGURES

FIGURE NUMBER	TITLE	PAGE NO
1	PHARMACEUTICAL CLUSTERS IN INDIA	15
2	SWOT ANALYSIS OF PHARMACEUTICAL INDUSTRY	16
3	MAJOR PLAYERS OF ONLINE PHARMACIES IN INDIA	17
4	SAMPLE SIZE CALULATION	29
5	RESEARCH MODEL	30
6	REALIABILITY AND VALIDITY TEST	33
7	CROSS LOADING	34
8	GENDER PROPORTION	37
9	ADOPTION OF ONLINE PHARMACIE WITH RESPECT TO AGE	38
10	ADOPTION OF ONLINE PHARMACIES EITH RESPECT TO INCOME	38
11	MEDICINES PURCHASED MONTHLY	39
12	BUBBLE GRAPH FORWHOM MEDICINES BEING PURCHASED	39
13	PURCHASE BASED ON INCOME	40
14	MEDICINES PURCHASE WITH RESPECT TO OCCUPATION	40
15	INTION TO ADOPT ONLINE PHARMACIES	41
16	ANN REPRESENTATION	45

17	IMPORTANCE USING ANN ALGORITHM	46
18	INTENTION TO ADOPT -GENDER WISE	47
19	DECISION TREE FOR ADOPTION INTENTION	49
20	BAR GRAPH FOR FREQUENTLY USED WORD	51
21	WORD CLOUD OF ADOPTION INTENTION	51
22	OPINION ON ONLINE PHARAMACY	51
23	SENTIMENTAL ANALYSIS FOR ADOPTION INTENTION OF ONLINE PHARMACIES	52

TABLE OF CONTENTS

Chapter	CONTENTS	PAGE NO
I	INTRODUCTION	10
1.1	ABOUT THE STUDY	10
1.2	PROBLEM STATEMENT	12
1.3	SCOPE AND SIGNIFICANCE OF THE STUDY	12
1.4	LIMITATIONS OF THE STUDY	13
II	INDUSTRY PROFILE	14
2.1	BACKGROUND OF THE INDUSTRY	14
2.2	MARKET SIZE	17
2.3	MAJOR PLAYERS	17
2.4	GOVERNMENT POLICIES AND REGULATIONS	18
2.5	RECENT TRENDS IN THE INDUSTRY	19
2.6	CHALLENGES FACED BY THE INDUSTRY	20
III	REVIEW OF LITERATURE	22
3.1	RESEARCH QUESTIONS	22
3.2	LITERATURE REVIEW	20
3.3	RESEARCH GAP	22
IV	OBJECTIVES OF THE STUDY	27
V	RESEARCH METHODOLOGY	28
5.1	RESEARCH DESIGN	28
5.2	SAMPLING DESIGN (population, sampling frame, sampling technique, sample size, sampling unit)	28

5.3	METHOD OF DATA COLLECTION	29
5.4	THEORETICAL FRAMEWORK	30
5.5	RESEARCH HYPOTHESIS	30
5.6	RESEARCH INSTRUMENT	35
5.7	PERIOD OF THE STUDY	35
5.8	TESTS OF RELIABILITY AND VALIDITY	35
5.9	STATISTICAL TOOLS USED FOR THE STUDY	36
VI	ANALYSIS AND INTEPRETATION	37
6.1	DESCRIPTIVE STATISTICS	37
6.2	ANALYSIS AND INTEPRETATION	41
VII	FINDINGS, SUGGESTIONS AND CONCLUSION	53
7.1	FINDINGS	53
7.2	SUGGESTIONS	55
7.3	CONCLUSION	56
VIII	BIBLIOGRAPHY	57
8.1	APPENDICES/ANNEXURE	57

ABSTRACT

This study attempts to discover and empirically validate the numerous characteristics that influence customer acceptance, usage, and intention to advocate using an online pharmacy to buy prescription drugs. Design/methodology/approach based on elements from well-established theoretical models, such as the extended unified theory of technology adoption and application. limits and implications of the study. The sample size for this study is geographically constrained. The study was carried out in the Indian region of Coimbatore. Real-world applications This study will offer intriguing insights for examining consumer adoption and usage intentions toward e-pharmacy on a scholarly level. This empirically validated analysis will provide managerial insights on the connections between the various constructs and consumer motivations for e-pharmacy uptake and usage intention. This study represented by the UTAUT 2 Model (Unified theory of acceptance and use of technology) is extension of original model utaut helps to understand consumer usage behavior.

Keywords: Online pharmacy, adoption, Behavioral Intention, Performance expectancy

I INTRODUCTION

1.1 ABOUT THE STUDY

Human life overall has been largely impacted by the digital transformation that is currently occurring. The rising use of internet technology has changed how daily activities are carried out. The use of the internet for shopping is widespread, and this includes purchasing medications. The development of online pharmacy has been signalled by this. The rise in the cost of prescription medications is primarily responsible for the emergence of online pharmacies, or e-pharmacies.

The convenience that internet pharmacies provide to their clients encourages them to frequently purchase medications through e- pharmacy. A key factor in determining whether clients use internet technologies to purchase drugs online is performance expectancy (Venkatesh et al. 2003). The ease with which clients can conduct their online shopping activities is referred to as the effort expectancy. The use of the internet to purchase anything, including medicines, is not without risk, though. Customers are cautious about the hazards involved with internet transactions. Therefore, it is acceptable for e-pharmacies to plan and carry out measures to advertise their businesses. These could involve actions where they use their social power to spread important information.

Aiming to better understand what consumers anticipate from online pharmacies and how the independent variables (performance expectancy, effort expectancy, social influence, facilitation control, hedonic motivation, habit, and price value) may affect dependent variable i.e., purchasing behaviors, this study takes these factors into account.

The main research issue, which has been established considering the previous backdrop, is "What are the numerous aspects that customers expect with respect to online pharmacies that influence purchase intention and decision?"

1.2 PROBLEM STATEMENT

There are various challenges faced by the customers who are purchasing online though many people out there using online to purchase groceries, foods etc., still there are many people are not coming forward to buy medicine through online, so here aim is to find the problems faced by customer to adopt medicines online.

- What are the various factors that customers expect with E-Pharmacies which leads to purchase intention and decision?

1.3 SCOPE AND SIGNIFICANCE OF THE STUDY

The goal of this study is to use the unified theory of acceptance and use of technology 2 (UTAUT2) model to analyse the variables that affect how long Coimbatore residents plan to continue using e-pharmacies, including performance expectations, effort expectations, social influence, facilitating conditions, price-saving orientation, hedonic, habit to behavioural intention, and behavioural intention to use behaviour.

Indian's e-commerce market will reach 95% of customer will buy products through online by 2040. Due to the increased investments made by e-commerce behemoths in the healthcare sector, the e-Pharmacy market trend is predicted to rise enormously at a CAGR of 17.3% in the ensuing six to seven years.

In the recent years Indian market has seen a steep upward curve in the number of net users and online shopping has become a trend. However, with the same rate the number of long term and chronic illness patients are increasing. Chronic diseases like diabetes, asthma, blood pressure, arthritis, thyroid etc., will be money spinner for pharmaceutical companies in about 5 years' time. medicines for such diseases usually don't have urgency as the intake as well planned these medicines are also expensive hence even 15 % discounts will make sense to the consumers.

This research will help us to look at the bigger picture as to which impact the customer to not to order medicines online.

1.3 LIMITATIONS OF THE STUDY

The limitation of the study would be time frame and cost factors, to collect data from the sample size that can be generalized. Moreover, the sample size is collected via convenience sampling process, because the non-probabilistic method, does not allow the study to be generalized to the whole Coimbatore population.

As e-pharmacy use grows in popularity, the impact of several moderating factors, including age, gender, and prior experience utilizing an online platform for Research should take drug purchases into account. Additionally, the online poll was only conducted in Coimbatore, India, placing a geographic restriction on the results, leading to the conclusion that they cannot be extrapolated to a broader demographic group. Convenience sampling was employed to obtain the data, which may not be the best sample strategy for determining how a technology platform would be adopted and used. For this study, the researchers advise combining qualitative and quantitative methods to increase the explanatory power of the suggested model testing of the

model's foundation Comparative research employing categorized consumer profiles may be an easy technique to expand the research and confirm the robustness of the suggested model.

Online medicine dealing is still in its development; on the one hand, its laws and regulations are not ideal, and its business model needs to be reinvented in light of the unique circumstances of the Indian medical distribution system; On the other hand, because health and medical products are tied to people's lives, they make people more cautious about buying them online, which reduces their desire to do so. This paper built an expanded model based on UTAUT 2 to analyze the effect variables and adoption intention of the online medicine purchase and collected data using scale questionnaires. The results of this study demonstrate which independent variable employed in this model is negatively significant for the decision to purchase medications online.

II INDUSTRY PROFILE

2.1 BACKGROUND OF THE INDUSTRY

Since the World Health Organization declared COVID-19 a global pandemic. Online pharmacies have become an increasingly popular option to obtain medication Owing to quarantine procedures implemented by the inter-agency task force (IATF) to limit the virus's spread.

In brief discussion online pharmacies are websites that promise to sell medicines to clients directly over Internet (Orazio, 2011). it includes both legitimate and illicit pharmacies. independent Internet only sites, online branches of “brick and mortar” pharmacist, and sites reflecting pharmacy collaborations are all included in definition of “Online pharmacies”.

According to Indian J pharmacol (2016) online pharmacies reduce transaction costs and the cost of obtaining pharmaceuticals through a more efficient centralised order processing system. the anonymity offered by the Internet encourages Customers to seek information about

medicines that they would otherwise avoid or asking their physician or at the offline pharmacies.

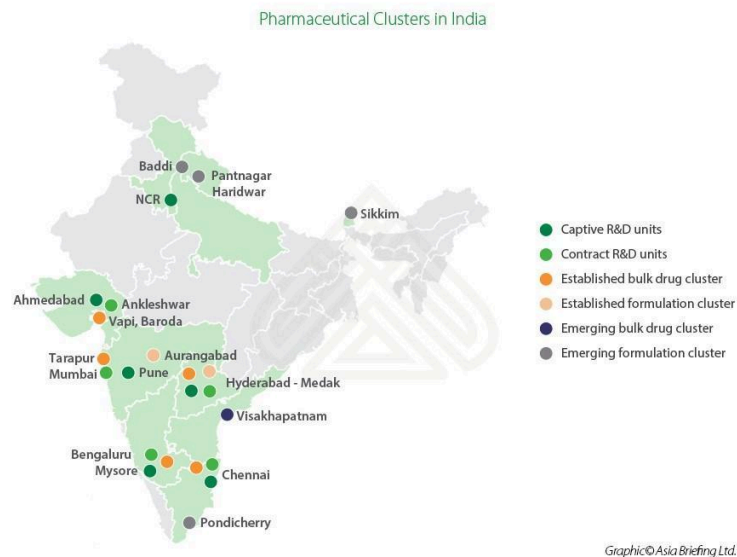


Fig 1. Pharmaceutical Clusters in India

consumers are increasing using Internet to not only find out information about their own health but also to self-diagnose and purchase health related products and services, where the reliability of information is doubtful due to changes in people's behaviour and responses due to the disease outbreak particularly the COVID-19 pandemic which has cost of global panic as a health stock (Orazio ,2009).

According to research 74 percentage of all individuals in us use the Internet and 61 percentage have looked for health or medical information on the Internet (fox and Jones 2009), furthermore 49 percentage have visited a website that gives information about a specific medicine ailment or problem.

Despite this penetration in Philippines is just 23.8% compared to 78.3% in North America as the result even if recent Nielsen Poll revealed significant advances access to Internet health information may remind limited.



Fig :2 Swot analysis of pharmaceutical industry

This Research is expected to gather results that concludes the perspective of users and patient- consumers in the usage of this online medicines.

2.2 MARKET SIZE

The size of the worldwide online pharmacy market was \$68,382.3 million in 2021, and it is projected to grow at a CAGR of 16.1% from 2021 to 2030 to reach \$261,684.5 million. This is attributed to the growing usage of e-prescriptions in hospitals and other healthcare facilities as well as the expansion of online services and the growing number of internet users.

India is one of the major countries for the supply of generic drugs, which account for around 20% of the total world supply in terms of quantity and for about 60% of vaccines. India's pharmaceutical industry currently accounts for approximately \$42 billion in global sales and is growing at a rate of 17% per year. According to India Ratings and Research, the country's pharmaceutical industry would have above 12% growth in the FY 2022–2023 period.

2.3 MAJOR PLAYERS

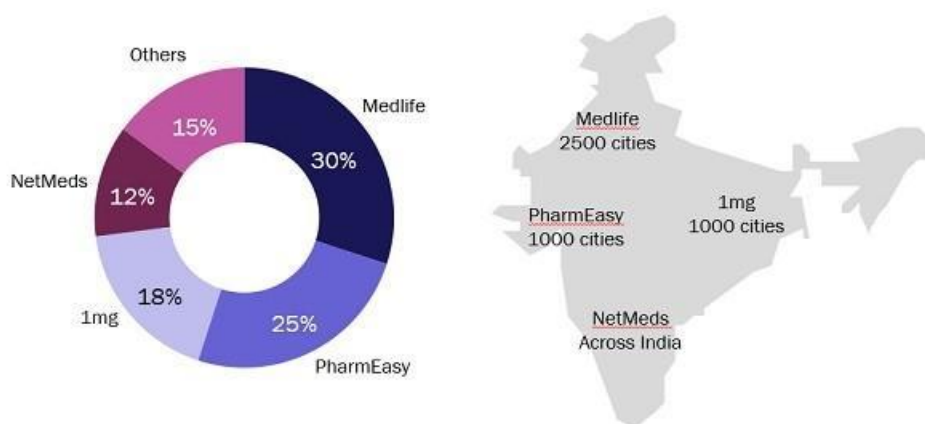


Fig 3: Major Players of Online pharmacies in India.

2.4 GOVERNMENT POLICIES AND REGULATIONS

The Drugs and Cosmetics Act, 1940 (D&C Act), Drugs and Cosmetics Rules, 1945 (D&C Rule), Pharmacy Act, 1948, Information Technology Act, 2000 (IT Act), Indian Medical Act, 1956 and Code of Ethics Regulations, 2002, Narcotic Drug and Psychotropic Substances Act, 1985, and Drugs and Magic Remedies (Objectionable Advertisement) Act, 1954 all govern the manufacture and sale of medicines in India.

The Draft Rules include several rules to Part VIB for the selling of drugs by online pharmacies. The word "e-pharmacy" has been added to the draught regulations to describe it as

the business of distributing or selling, stocking, exhibiting, or making medications available for sale through a web portal or other electronic form. Additionally, the phrases "e-pharmacy portal" and "selling via e-pharmacy" have been appropriately defined.

In order to monitor e-pharmacies, a complaint redressal mechanism has been put in place that gives consumers the option of reporting any suspicion of the supply of subpar quality, adulterated, or misbranded drugs through e-pharmacies to the state drugs controller (the "Drugs Controller") in addition to the Consumer Protection Act of 1986. The Draft Rules, however, are still awaiting approval.

2.5 RECENT TRENDS IN THE INDUSTRY

- Artificial Intelligence:

Most sectors are embracing artificial intelligence. In place of human intellect, computer systems are used to carry out tasks. It comprises decision-making, voice recognition, and vision perception. Translation between many languages is also a part of it.

Increasing the usage of AI in the pharmaceutical industry may, among other things, speed up the research and development of new drugs. A few applications for AI include clinical trials, drug research, and fraud detection.

- Blockchain

Blockchain technology has proven successful in the financial sector, and many have begun to comprehend its applications, including those in the pharmaceutical sector. A key technology for the distribution and creation of drugs is blockchain. Blockchain primarily serves the objective of streamlining transactions while improving security and transparency and doing so without a middleman.

- Digital Marketing

The main healthcare systems are now driven by digital technology. They have policies and goals for digital health, have built technological infrastructure, and use data collected by the system to assess results and develop new initiatives. It had already migrated into the supply chain and the B2B world before the epidemic.

Prior to 2020, digital marketing was a weakness for many pharmaceutical businesses. Healthcare organizations cannot risk not having an internet presence in 2021. As a result, the pharmaceutical sector considers digital marketing to be a key trend right now.

- Big data and Analytics

As the pharmaceutical sector expands, analysis and data are required. A lot of information is gathered since there is a wide range of medications and their development to investigate. This indicates that sophisticated software is required to examine it and determine a certain value.

The sectors have moved toward digitalization and are willing to delegate data management tasks to outside parties, making it a crucial component of the pharmaceutical sector. Modern procedures and technology have advanced, making

analysis easy and significant. The analysis now has diagnostic, descriptive, predictive, and prescriptive capabilities.

- Research and Development

The importance of drugs is receiving more attention. Pharmaceutical firms make sure that research and development focuses on the desired outcome. According to the requirements of the patients, an attempt is made to attain efficacy and efficiency.

Research has been done explicitly to identify drugs that enhance life quality and experience in a significant way. Pharmaceutical businesses are now reviewing their research and development processes to make sure the procedures are honed and targeted as a result.

2.6 CHALLENGES FACED BY THE INDUSTRY

- Demand Forecasting

Creating market predictions and raising customer service standards is the pharmaceutical industry's participants' toughest problem. For pharmaceutical firms to stay competitive and balance supply and demand, demand forecasting is a crucial component. Pharma firms face a significant difficulty in ensuring that all commodities are manufactured at the right time and delivered in an efficient manner.

- Price Fluctuation assessment

When it comes to healthcare analytics and other data analytics in the healthcare industry developments, recent study of the pharmaceutical business reveals that many major

firms in the sector face challenging times. In order to increase profit margins, the pharmaceutical business has recently needed adequate methods for analyzing price structures. One of the main difficulties facing the pharmaceutical sector is analyzing the altering consumer behavior and fluctuating costs.

- Risk Management

It is crucial to control hazards in the systems for quality control and medication manufacture. It becomes crucial for businesses in the pharmaceutical sector to ensure product quality. As a result, it is difficult for pharmaceutical businesses to pinpoint potential dangers related to a product or the procedures used in its research, production, and distribution.

III REVIEW OF LITERATURE

3.1 RESEARCH QUESTIONS

1.How does Performance Expectancy influence Behavioral Intention of online-pharmacies adoption?

2. How does Effort Expectancy influence Behavioral Intention of online-pharmacies adoption?
3. How does Social Influence Behavioral Intention of online-pharmacies adoption?
4. How does Facilitating condition influence Behavioral Intention of online-pharmacies adoption?
5. How does Facilitating Condition influence Use Behavior of online-pharmacies adoption?
6. How does Price Saving Influence Behavioral Intention of online-pharmacies adoption?
7. How does Hedonic influence Behavioral Intention of online-pharmacies adoption?
8. How does Habit influence Behavioral Intention of online-pharmacies adoption?
9. How does Behavioral Intention influence Use Behavior of online pharmacies?

3.2 LITERATURE REVIEW

(Ma, 2021) the desire of non-adopters to use online pharmacies. This study further incorporated trustworthiness and perceived risk based on the technology acceptance paradigm. It was discovered that non-adopters' intention was greatly influenced by how valuable and trustworthy they felt.

(Sezgin et al., 2014) The goal of the study is to evaluate and identify the elements that affect pharmacists' acceptance of the system. In this ongoing investigation, a brand-new research model (P-TAM) was created, and 2169 pharmacists from various areas of Turkey participated in a survey. Statistical investigations were carried out to evaluate the model's robustness and dependability.

(Srivastava & Raina, 2020) This study attempts to discover and experimentally evaluate the numerous characteristics that influence customer acceptance, usage, and intention to advocate using an online pharmacy to buy prescription drugs. Design/methodology/approach: A model was presented for the study that is based on concepts from well-known theoretical models, including the extended unified theory of acceptance and use of technology, the technology acceptance model, and self-determination theory. A sample size of 184 respondents was used to verify the model, and links between the different discovered components were established and confirmed using the partial least squares approach and factor analysis.

(Bianchi & Andrews, 2012) It seeks to respond to demands for research on consumers' post-adoption acceptance of a technology to learn more about the variables most important in predicting continuance behavior. Design/methodology/approach: The study examines a model of how customers' attitudes and intentions to make more online purchases are affected by their perceptions of risk and trust. The technique employed is an online survey. To analyze the data, structural equation modelling was used (SEM). Findings: The investigation shows that attitude has a favorable impact on intentions to continue buying and that attitude has an inverse association with perceived danger online.

(Tarhini et al., 2016) Numerous studies have demonstrated that the adoption of internet banking (IB) is influenced not just by banks' or governments' backing, but also by IB users' views and experiences. Academics in industrialized nations have demonstrated good achievements in IB studies. However, nothing is known on IB user adoption in Lebanon. The goal of this essay is to investigate the elements that may help or impede IB usage and acceptability in Lebanon.

(Rahi et al., 2018a) The goal of this study is to examine how the unified theory of acceptance and use of technology (UTAUT) functions in the context of online banking uptake. 398 internet banking customers' information was gathered using a quantitative approach-based survey. The structural equation model (SEM) technique was employed for statistical analysis. According to the study's findings, the UTAUT model offered a solid theoretical base for research into technology adoption.

(Yin et al., 2016) The purpose of this article is to analyze customer acceptability of online pharmacies and to identify the many factors that influence the desire to buy medicines online. The unified theory of acceptance and application of technology (UTAUT) model serves as the foundation for the five explanatory constructs in this study. In China, consumer survey data (n=274) were gathered, and the results were compared to the suggested research methodology.

(Gu et al., 2021) For the analysis in this work, partial least square structural equation modelling was applied. The findings show that factors such as expected effort, social influence, favorable circumstances, task-technology fit, trust, privacy, and individual information technology innovation play a major role in determining whether or not people would embrace electronic health technology.

(Owusu Kwateng et al., 2019) The goal of this article is to explore, using the unified theory of acceptance and use of technology 2 (UTAUT2) model with age, education, user experience, and gender as moderators, variables that affect consumers to embrace and subsequently utilize

m-banking services in Ghana. Design/methodology/approach: The study chose 300 Ghanaian users of m-banking services as respondents using a questionnaire survey.

(Fatima et al., n.d.) The purpose of this study is to describe how people purchase medications online. The first goal of the article was to identify the variables that affect pharmaceutical purchases made online. Second, it went on to research the most important aspects of the same. The goal of the study is to determine how customer purchasing habits for generic and brand-name medications vary.

(Naveed et al., 2020) The goal of the current study is to determine what influences Saudi university students' adoption of mobile learning. The technology acceptance model (TAM) and the unified theory of acceptance and use of technology (UTAUT) models served as the foundation for the development of a theoretical model of mobile learning acceptance. Five distinct constructs were chosen as the ones that, according to theory, contributed the most to the use of mobile learning and were empirically examined.

(Sabbir,2020) determining the elements that influence client adoption of Internet banking services is necessary. A thorough research model was created and scientifically tested based on the UTAUT model, the IS quality components, and the idea of trust.

(Walton et al., n.d.) intends to research technology acceptance and usage in a consumer setting by expanding the unified theory of acceptance and use of technology (UTAUT). Hedonic motivation, price value, and habit are three constructs that our revised UTAUT2 includes. The

impacts of these dimensions on behavioral intention and technology usage are predicted to be moderated by individual characteristics, including age, gender, and experience.

(ONLINE PHARMACY -2, n.d.) This study looks at how internet pharmacies are being used in India, a developing nation. To do this, the Technology Acceptance Model (TAM) was expanded to incorporate Trust and Perceived Risk. This resulted in the development of a conceptual model. Using AMOS software and structural equation modelling (SEM), the conceptual model was put to the test empirically. The primary variables that affect customers' attitudes and behavioral intentions regarding the adoption of online pharmacies in India, according to the study's findings, are trust and perceived usefulness.

(Haikal et al., 2022) The goal of this study was to identify the variables that affected how acceptably e-health applications were received. The most crucial elements, such the application's success rate and better e-health application performance, were revealed by the study's findings. The research paradigm employed was the Unified Theory of Acceptance and Use of Technology 2 (UTAUT 2).

(Yuan-Fang Song,2018) to identify the factors that determine customer acceptance of Internet banking services. Based on the UTAUT model, the IS quality constructs, and the concept of trust, a comprehensive research model was developed and empirically examined and found the impact of usage of online banking system.

(Abu Bakar, 2019) In order to assess consumer acceptance of omni-channel usage, this article attempts to establish a theoretical framework that integrates the Unified Theory of Acceptance

and Use of Technology (UTAUT2), Personal Innovativeness, brand image, and fashion engagement. The research paradigm employed was the Unified Theory of Acceptance and Use of Technology 2 (UTAUT 2).

3.3 RESEARCH GAP

Online pharmacies have offered discounts and fast delivery services at consumers' and but doesn't concentrate on promotions and advertisements (sei Atiemo, K. A., & Appiah, C. ,2019).In my further research ,promotion on online-pharmacies would trigger the increased popularity of online pharmacies around , this might influence buyers to buy medicines online. The target population of this study is concentrated only in Coimbatore, and who are more likely to adopt and exploit the benefits of any technological changes and in my further research I've Planned to use classification and regression algorithms to predict the behavioral intention of customer to buy medicines online.

IV OBJECTIVES OF THE STUDY

The purpose of this research is to examine the adoption of online pharmacies in India. Understanding the factors that affect usage behavior of online pharmacies can develop marketing and supply chain strategies to ensure that people use this new service, And This study attempts to discover and experimentally evaluate the numerous characteristics that influence customer acceptance, usage, and intention to advocate using an online pharmacy to buy medications.

The objective for the study is to measure as follows:

- To examine the demographics characteristics of the respondents.
- To identify the factors referring to the intention to buy medicines online.
- To examine the factors that affects customers to purchase medicines online.
- To Predict the intention of buying medicines through online among general population

V RESEARCH METHODOLOGY

5.1 RESEARCH DESIGN

This study will use a quantitative cross-sectional research strategy, as outlined by Bryman and Bell (2015). The primary data will be collected through a web-based survey administered to a sample of internet users. The sampling process will be non-probabilistic, utilizing convenience and snowball sampling techniques, using social media platforms such as Facebook, Instagram, and WhatsApp, and mall-intercepts by distributing self-administered questionnaires. Respondents will be asked to share the survey with their social friends and also circulated forms through medicals the people who purchased medicines to increase participation.

5.2 SAMPLING DESIGN

The target respondents for our survey are the general population actively using smart phones, residing Coimbatore.

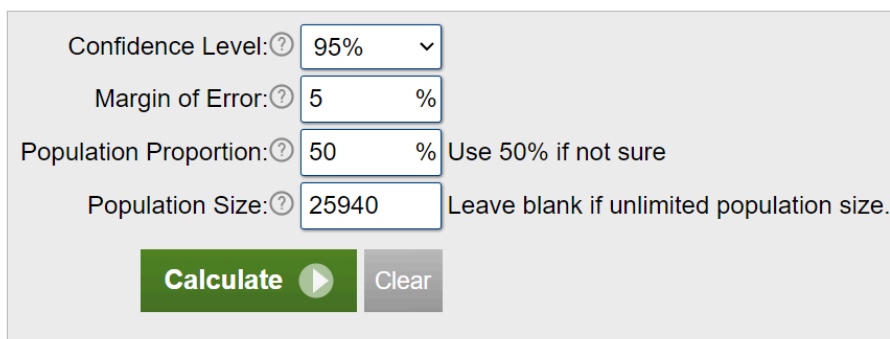
The sampling criterion for this study was convenience sampling, which is a type of

non-probability sampling method which is considered due to time constrain, where the sample is taken from a group of people who are purchasing through online and are easy to contact or to reach (Tairan, N. ,2020)

Result

Sample size: **379**

This means 379 or more measurements/surveys are needed to have a confidence level of 95% that the real value is within $\pm 5\%$ of the measured/surveyed value.



The screenshot shows a web-based sample size calculation tool. It features four input fields: 'Confidence Level' set to 95%, 'Margin of Error' set to 5%, 'Population Proportion' set to 50%, and 'Population Size' set to 25940. Each field has a help icon (question mark in a circle). To the right of the 'Population Proportion' field is the text 'Use 50% if not sure', and to the right of the 'Population Size' field is the text 'Leave blank if unlimited population size.' At the bottom, there are two buttons: a green 'Calculate' button with a play icon and a grey 'Clear' button.

Confidence Level: ?	95%	▼
Margin of Error: ?	5	%
Population Proportion: ?	50	% Use 50% if not sure
Population Size: ?	25940	Leave blank if unlimited population size.

Calculate ▶ **Clear**

Fig 4. Sample size calculation

Where SS stands for Sample Size.

Z is 1.96 (95% confidence level);

P is prevalence level (required sample size is 0.5);

Q is (1 p); and E is error term (0.05).

5.3 METHODS OF DATA COLLECTION

Primary data method was followed, where the data was collected by providing questionnaires via online Forms to reach out to many respondents due to time constraints the survey is conducted only through online mode. So, the data collected is the primary data

collected with a structured questionnaire where it is sent to the respondents through an online platform and circulating the forms in person.

5.4 THEORITICAL FRAMEWORK

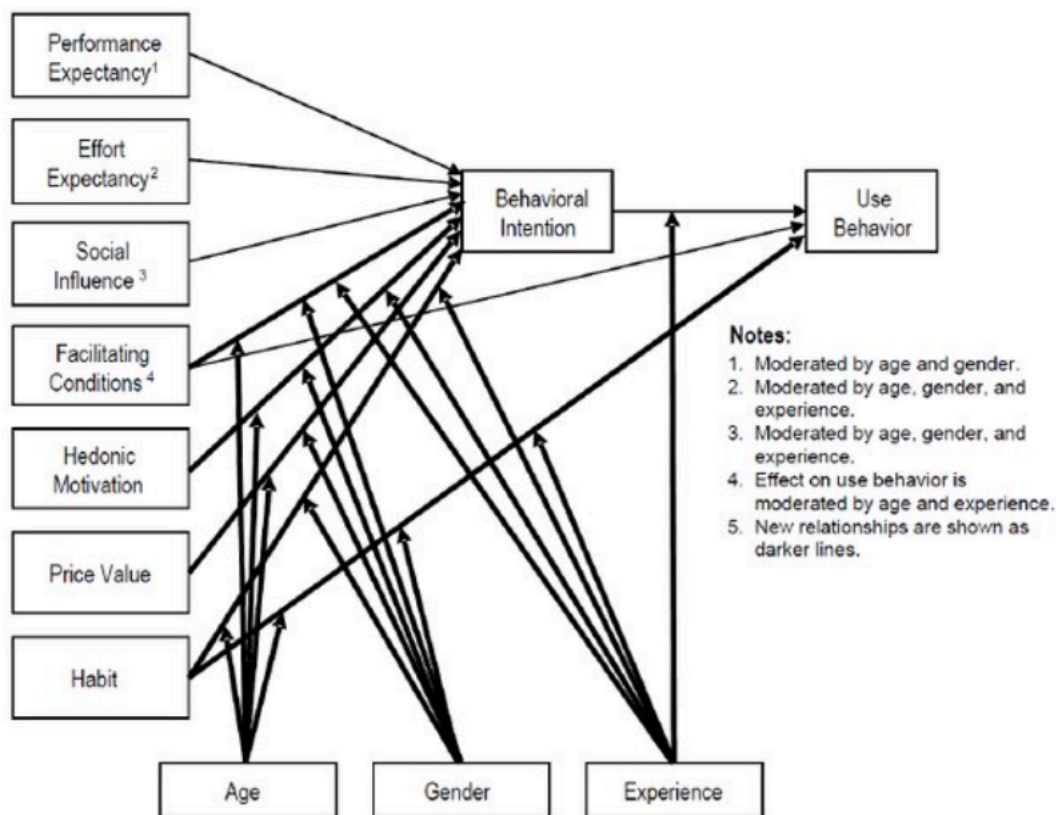


Fig 5. Research Model: UTAUT2
(Venkatesh et.al.2012)

5.5 RESEARCH HYPOTHESIS

Performance expectancy:

PE is "the degree to which the user may trust that employing the technology would allow him or her to overcome difficulties and help him or her to reach the intended objective in work performance," according to Venkatesh et al. If the system is successful, it will encourage the use of services (Tojib & Tsarenko, 2012).

Many earlier health studies demonstrate a positive link between PE and BI, according to previous research (Alam et al., 2019, 2020). PE has an impact on users' intentions to utilize a new e-pharmacy service that can promote knowledge about the general health system (Gagnon et al., 2016). Slade et al. (2013) claim that PE aids in the development of satisfying health experiences. PE influences people's willingness to accept new technologies in a favorable way (Khan et al., 2018). The supporting evidence supports the following hypothesis, which is developed:

H1: PE is correlated with patients' BI to utilize e-pharmacies in a good way.

Effort expectancy:

This is defined as the consumer's ease of use of technology (Venkatesh et al., 2012). In the context of online pharmacy, effort expectancy is understood as consumers' perceived ease of using online pharmacy for purchasing medications. Prior studies have claimed that for a new technology the ease of use is no longer a barrier; as users are technologically sound (Wang et al., 2014). This thereby means that effort expectancy does influence consumer behavioral intention toward online pharmacy positively, thus the following hypothesis:

H2. Effort expectancy has a positive effect on behavioral intention to adopt online pharmacy to purchase medicines.

Social Influence:

When it comes to adopting information technology, people are undoubtedly influenced by others in their social networks and communities, such as coworkers, friends, and family members. The impact is known as SI in the area of technology. SI is defined as "the extent to which a person perceives that other significant [people] feel that he or she should utilise the new system" (Venkatesh et al., 2003). SI has been properly considered and validated as a predictor of personal conduct (Zuiderwijk et al., 2015). The author of a study done in South Korea discovered that SI had a considerable impact and noted that it is a crucial precedent for BI (Kim et al., 2011). The favorable impact of SI on users' intentions to utilize e-pharmacies services is also supported and explained by previously published studies (Ahmad & Khalid, 2017; Sun et al., 2013). This study provides empirical evidence for the close connection between SI and BI. Consequently, we arrive to the following conclusion:

H3: The usage of e-pharmacies by patients is favorably correlated with SI.

Hedonic Motivation:

This is the joy a user experiences while implementing or utilizing a new platform or technology (Venkatesh et al., 2012). Hedonic motivation directly affects the user's desire to accept technology in various circumstances, according to prior study findings. Therefore, we suggest the following:

use of online pharmacies

H4. Hedonistic incentive has a favorable impact on the behavioral intent to use an online pharmacy to buy medications.

Facilitating condition:

FC is a major additional Online-pharmacies technology predictor. BI and the adoption of technology were demonstrated to be significantly and favorably impacted by FC in prior study. "The degree to which a person trusts that a structural and functional infrastructure exists to facilitate the usage of the system" is how FC are defined (Venkatesh et al., 2012). To create the intention to use, a user must have access to a sufficient collection of FC, such as hardware/software, technical assistance, online tutorials, and online demos. Numerous studies have demonstrated the positive impact of FC on how people use e-pharmacies services and HI systems (Boontarig et al., 2012). The following is our hypotheses:

H5: FC are connected with customers' BI to a positive degree.

Price Value:

The trade-off between the price spent for utilizing the technology and the deemed advantages obtained is known as price value in the UTAT2 (Dodds, Monroe and Grewal, 1991; Venkatesh et al., 2012). This covers the costs associated with buying or using the service, such as equipment, software, data, and other fees (Wei et al., 2009). According to Zhou and Wong (2004), one of the most important factors influencing buying impulsiveness is price. According to Monsuwe et al. (2004), savings from online buying is a key reason customers make purchases online, hence pricing is an essential consideration.

H6: PV have favorable impact on behavioral intent to use online-pharmacies to buy medicines

Habit:

Habit is the degree to which people typically carry out behaviors instinctively (Limayem et al., 2007). According to research by Venkatesh et al. (2012), habit has a direct and indirect impact on one's inclination to utilize technology. The empirical impact of habit on purchasing intention has been studied (Khalifa and Liu, 2007; Chiu et al., 2012). Depending on the type of experience and how tough or simple it is for the customer to use, gaining more experience might have a good or negative impact on adoption (Venkatesh and Bala, 2008). Essentially, habits are created by recurrent encounters.

H7: Habit of using e-pharmacies have positive correlation towards adoption of purchasing medicines online.

Behavioral Intention:

BI provides information on a person's preparedness to engage in a particular behaviour and is seen as an immediate antecedent of usage behaviour. Actual behaviour is defined as "the evident, observable reaction with regard to a particular target in a given context" (Ajzen, 1991). In information technology acceptance studies, there is a lot of evidence that BI has a major impact on AU (e.g., Venkatesh et al., 2003, 2012; Venkatesh and Zhang, 2010; Tarhini et al., 2013b). The IB context has only lately been included as well (Im et al., 2011; Ma et al., 2010; Yu, 2012). In order to investigate the following claim, the present study:

H8. Customers' BI will significantly and positively affect how they actually use the Online pharmacies.

Age, gender, education:

Age, gender, and education effects in order to understand and validate if these variables did have an influence on the behavioral intention to adopt online pharmacies for the purchase of medications, the researchers attempted to establish a hypothesis connected to age, gender, and education for the demographic characteristics. As a result, we propose the following hypothesis:

As a result, we propose the following hypothesis:

H9: Age positively increases behavioral intention to embrace online pharmacies for medication purchases.

H10: Gender effects behavioral intention to use an online pharmacy to buy medications favorably.

H11: Education impacts behavioral intention to use an online pharmacy to buy medications positively.

5.6 RESEARCH INSTRUMENT

The data will be collected by circulating google forms and the analysis will be done using machine learning algorithms. The research instrument used for the survey is a standardized, structured questions for the customers who use online for purchasing medicines. The survey form is the main source for the primary data collection. Machine learning algorithm will be used for predicting the behavioral intention of the customers who buys medicines online.

5.7 PERIOD OF THE STUDY

The study was conducted for the period of 6 months from Nov 2023 to May 2023.

5.8 TESTS OF RELIABILITY AND VALIDITY

RELIABILITY TEST:

Classic reliability coeffs.									
	PE	EE	SI	FC	HM	H	PV	PSQ	BI
Composite reliability	0.884	0.890	0.874	0.862	0.901	0.826	0.894	0.814	0.842
Cronbach's alpha	0.738	0.835	0.806	0.757	0.834	0.683	0.764	0.713	0.719
Additional reliability coeffs.									

VALIDITY TEST:

	PE	EE	SI	FC	HM	H	PV	PSQ	BI
PE	0.890	0.734	0.734	0.647	0.087	0.316	0.443	0.444	0.596
EE	0.734	0.819	0.751	0.664	0.125	0.544	0.574	0.554	0.682
SI	0.734	0.751	0.796	0.686	0.233	0.431	0.424	0.454	0.582
FC	0.647	0.664	0.686	0.822	0.162	0.558	0.365	0.585	0.688
HM	0.087	0.125	0.233	0.162	0.867	0.139	0.421	0.431	0.239
H	0.316	0.544	0.431	0.558	0.139	0.783	0.260	0.482	0.602
PV	0.443	0.574	0.424	0.365	0.421	0.260	0.899	0.531	0.405
PSQ	0.444	0.554	0.454	0.585	0.431	0.482	0.531	0.688	0.557
BI	0.596	0.682	0.582	0.688	0.239	0.602	0.405	0.557	0.800

Fig 6. Reliability and validity test

All factor loadings are significant. Cronbach's alpha; AVE = Average variance extracted; CR = Composite reliability.

From Fig.6 Reliability and validity test done. I could conclude that composite reliability and Cronbach's alpha are 0.7 or higher is considered good for further research.

	PE	EE	SI	FC	HM	H	PV	nolname	BI	Type (as defined)	SE	P value
PE1	(0.890)	-0.064	-0.039	0.191	-0.073	0.208	-0.006	-0.015	-0.125	Reflective	0.106	<0.001
PE3	(0.890)	0.064	0.039	-0.191	0.073	-0.208	0.006	0.015	0.125	Reflective	0.106	<0.001
EE1	0.303	(0.867)	-0.044	-0.103	0.073	0.062	-0.081	-0.295	0.194	Reflective	0.107	<0.001
EE2	0.064	(0.860)	-0.116	0.041	-0.019	0.003	0.073	0.218	-0.037	Reflective	0.108	<0.001
EE3	-0.262	(0.769)	0.314	0.046	0.007	0.075	-0.066	-0.138	-0.395	Reflective	0.112	<0.001
EE4	-0.151	(0.774)	-0.135	0.024	-0.067	-0.148	0.074	0.225	0.216	Reflective	0.112	<0.001
SI1	0.396	-0.353	(0.833)	0.112	-0.091	-0.049	-0.056	-0.126	0.432	Reflective	0.109	<0.001
SI2	-0.062	0.102	(0.852)	-0.072	0.066	-0.267	-0.112	0.138	0.198	Reflective	0.108	<0.001
SI3	0.111	-0.194	(0.740)	-0.046	-0.274	0.267	0.212	0.116	-0.444	Reflective	0.113	<0.001
SI5	-0.476	0.464	(0.755)	0.002	0.295	0.093	-0.020	-0.130	-0.265	Reflective	0.112	<0.001
FC1	0.137	-0.033	0.165	(0.884)	-0.030	0.049	-0.082	-0.223	0.145	Reflective	0.107	<0.001
FC2	-0.043	0.277	-0.082	(0.854)	-0.152	-0.114	-0.005	0.074	0.331	Reflective	0.108	<0.001
FC3	-0.116	-0.288	-0.105	(0.721)	0.217	0.075	0.107	0.186	-0.571	Reflective	0.114	<0.001
HM3	0.309	0.307	-0.478	0.193	(0.839)	-0.131	-0.004	-0.278	-0.173	Reflective	0.109	<0.001
HM4	-0.092	-0.559	0.384	-0.067	(0.859)	0.177	0.183	0.078	0.244	Reflective	0.108	<0.001
HM5	-0.201	0.247	0.078	-0.116	(0.900)	-0.047	-0.171	0.185	-0.072	Reflective	0.106	<0.001
H1	0.642	-0.360	-0.057	-0.147	0.033	(0.787)	0.005	-0.172	0.003	Reflective	0.111	<0.001
H2	-0.105	-0.186	0.062	0.156	-0.156	(0.801)	0.381	-0.129	0.044	Reflective	0.110	<0.001
H3	-0.555	0.570	-0.006	-0.013	0.130	(0.759)	-0.408	0.315	-0.049	Reflective	0.112	<0.001
PV3	-0.016	0.046	-0.164	0.093	0.011	-0.027	(0.899)	-0.006	-0.042	Reflective	0.106	<0.001
PV4	0.016	-0.046	0.164	-0.093	-0.011	0.027	(0.899)	0.006	0.042	Reflective	0.106	<0.001
PSQ2	0.386	-0.026	-0.214	0.256	0.547	0.270	-0.148	(0.770)	-0.355	Reflective	0.112	<0.001
PSQ3	-0.424	-0.271	0.180	0.322	-0.422	-0.038	0.224	(0.745)	0.081	Reflective	0.113	<0.001
PSQ4	0.023	0.276	0.039	-0.543	-0.133	-0.223	-0.066	(0.804)	0.265	Reflective	0.110	<0.001
BI1	0.765	0.011	-0.200	-0.006	0.031	0.256	-0.227	0.132	(0.765)	Reflective	0.112	<0.001

Fig 7. Combined loadings and cross loadings

From fig7. I could conclude that all items that are associated are strongly associated with the factor in its loading.

5.9 STATISTICAL TOOLS USED FOR THE STUDY

- SPSS
- R
- TABLEAU

VI ANALYSIS AND INTEPRETATION

6.1 DESCRIPTIVE STATISTICS

Participant characteristics

382 successful interviews were completed by using forms shared over to various respondents in person. About 60.7% of female participants in the sample and the male respondents where about 39.21%.

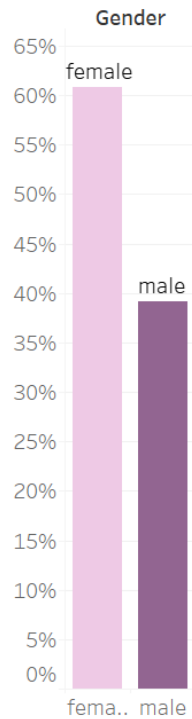


Fig 8. Gender proportion

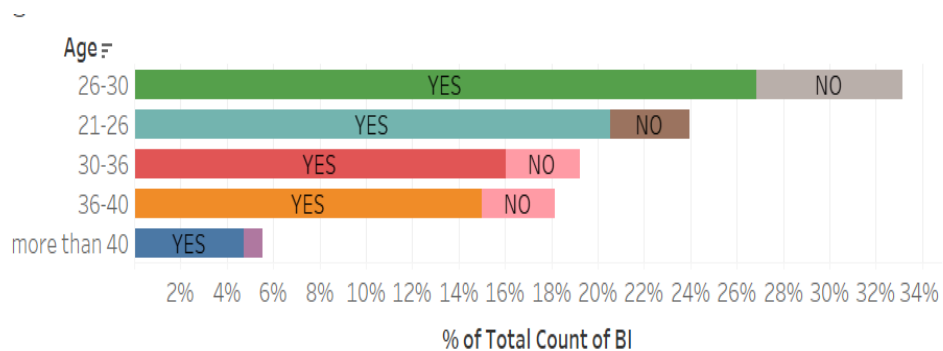


Fig 9. Adoption intention of e-pharmacy-Age wise

Around 26.8% where 26-30 age grouped people who are interested to adopt online pharmacy, among 21-26 Age grouped people are 20.5% who are considering adopting online pharmacies,

30-36 Age grouped shown interest to adopt online pharmacies upto 16.6% and 15.3% of people are 36-40 aged people who are considering to adopt online pharmacies and Others are who are more than the age of 40 are preferring less than 5%.

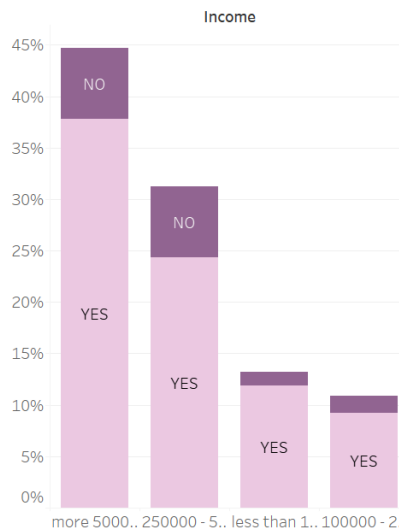


Fig 10. Adoption intention of Online pharmacy -Income wise

The people who are earning more than 500000 LPA have shown interest to buy medicine through online and considering online pharmacies as useful application.24.3% were the people who are earning in the range of 250000-500000.

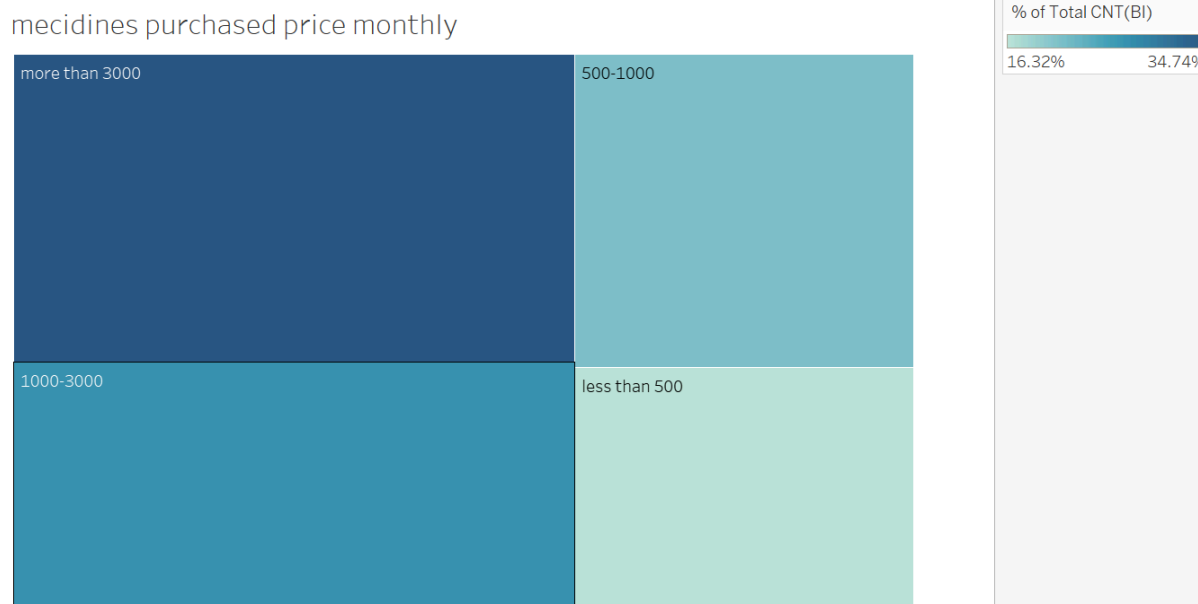


Fig 11. Medicines purchased monthly.

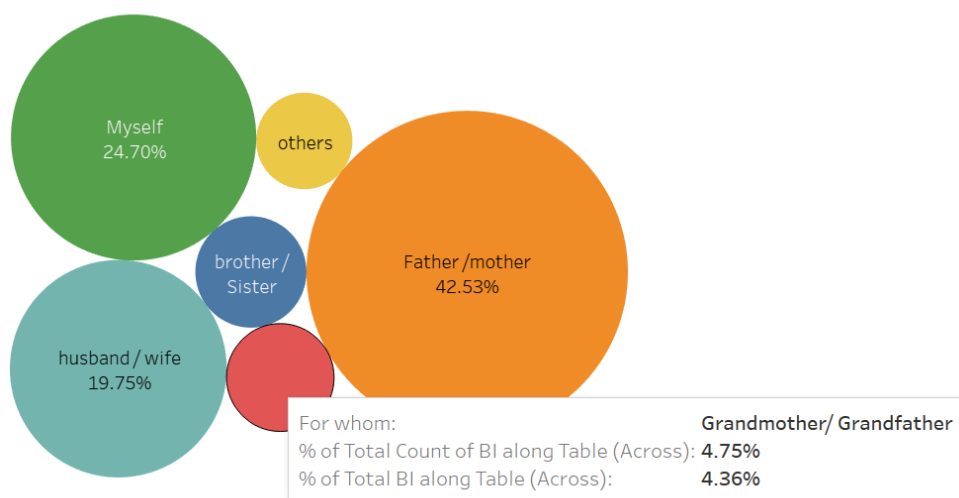


Fig 12. Medicines Purchased for Whom

The people who are purchasing medicines frequently are buying medicines for their father/mother ie.42% of people who are coming to pharmacies are buying medicines And 24.7% of people who are buying medicines for themselves and 19.7% are buying medicines for their husband /wife.

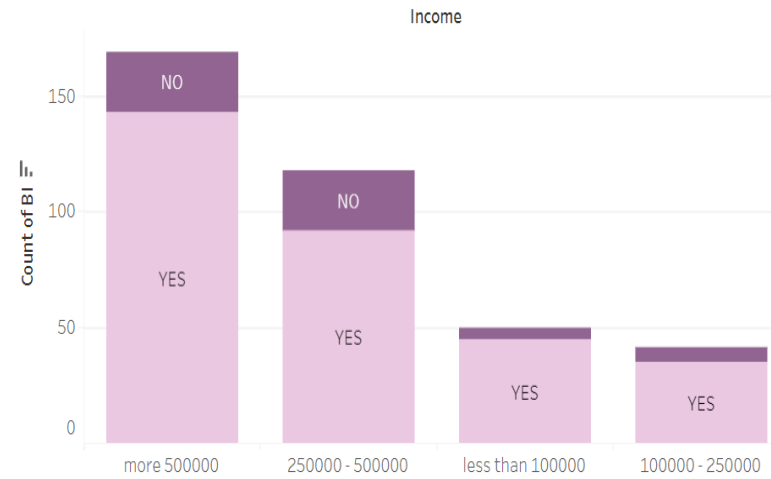


Fig 13. Purchase based on income.

With respect to (Fig:13) majority of respondents who have the annual income more than 5LPA have shown interest to adopt online pharmacies, 25% of respondents who are earning 250000-500000 have shown interest to adopt online pharmacies.



Fig 14. Medicines purchased based on occupation.

With respect to fig13 we could conclude that the Respondents who are working In IT field are really interested in buying medicines online.

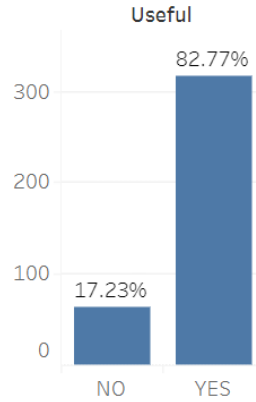


Fig 15. Intention to adopt online pharmacies.

From Fig14, we could say that 82.77% of the total respondents from the survey have voted that they have intention to adopt online pharmacies whereas 17.23% told that they have no intent to adopt online pharmacies.

6.2 ANALYSIS AND INTEPRETATION

6.2.1 Factors associated with the adoption intention of online pharmacy with respect to ANN.

Classification

ANALYSIS	TRAINING	TESTING
TEST 1	78.5%	85.0%
TEST 2	79.6%	81.9%
TEST 3	80.2%	83.6%
TEST 4	79.5%	81.5%
TEST 5	81.5%	78.6%

Table1.Multiple testing of ANN

Sample	Observed		Predicted	
		1	2	Percent Correct
TRAINING	1	216	0	100.0%
	2	59	0	0.0%
	Overall percent	100%	0.0%	78.5%
TESTING	1	85	0	100%
	2	15	0	0.0%
	Overall percent	100%	0.0%	85.0%

With the respect to (Table1) ANN algorithm was used and tried multiple times to get good accuracy. I could conclude that from the above table the overall percentage of accuracy was 85.0% which is higher than other test with Ann. So I could choose this analysis for my future interpretation.

Classification

Table 2. Accuracy of ANN algorithm

ANN algorithm or Artificial Neural Network is also used for this study's prediction. The accuracy level of the algorithm for adoption intention of online pharmacy is 85.0%. From the Table 2, we could interpret the importance of each independent variable with respect to the dependent variable (Adoption Intention of online pharmacy).

Independent Variable Importance

	IMPORTANCE	NORMALIZED IMPORTANCE
AGE	.053	66.5%
GENDER	.023	29.4%
MARITAL STATUS	0.53	41.7%
EDUCATIONAL LEVEL	0.52	64.9%
EXPERIENCE	0.63	78.7%
OCCUPATION	0.40	50.7%
FREQUENTLY_PURCHASED MEDICINES	0.54	67.5%
MONTHLY EXPENSE	0.32	39.9%
MONTHLY PURCHASE	0.45	55.3%
ILLNESS_MEDICINES	0.21	26.3%
PE	0.68	85.3%
EE	0.80	100%
SI	0.70	87.4%
FC	0.63	79.3%
HM	0.64	80.7%
H	0.66	82.8%
PV	0.58	73.0%
PSQ	0.61	77%

Table 3. Artificial Neural Network results for adoption intention of online pharmacy

The most important variable present is one of this ANN Algorithm Constructs which is the Effort expectancy which has 100% normalized importance followed by Social Influence to action which had an importance of 87.4% and the next place is taken by Performance Expectancy which had an importance of 85.3%.

We could say that the adoption intention is significantly proportional to all the variables present as all the variable has some importance with respect to the dependent variable. Here the Price value is the least important variable which means it has very less connection to intention to adopt online-pharmacies. The most important variables are Effort Expectancy, Performance Expectancy, Social Influence.

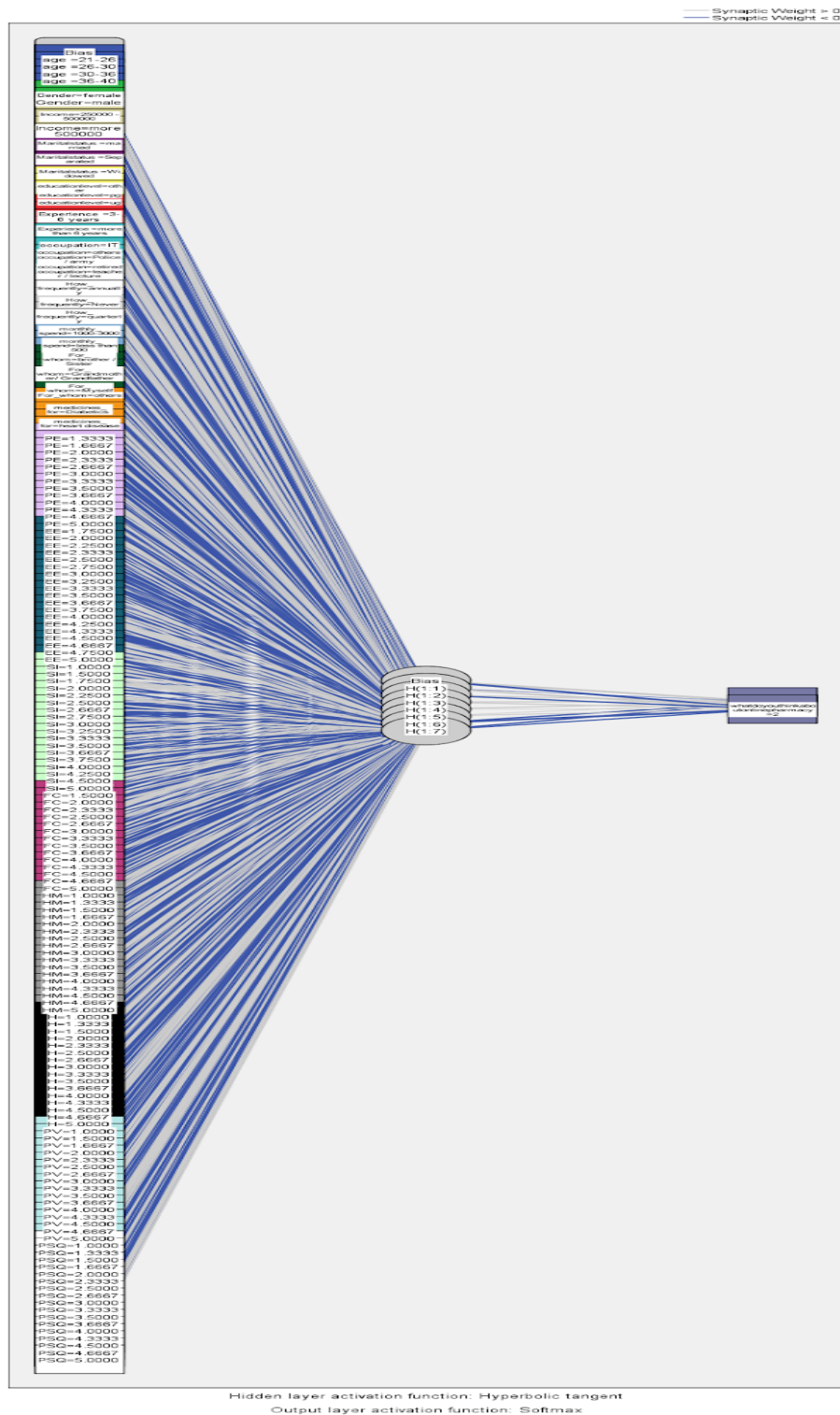


Fig 16. ANN representation

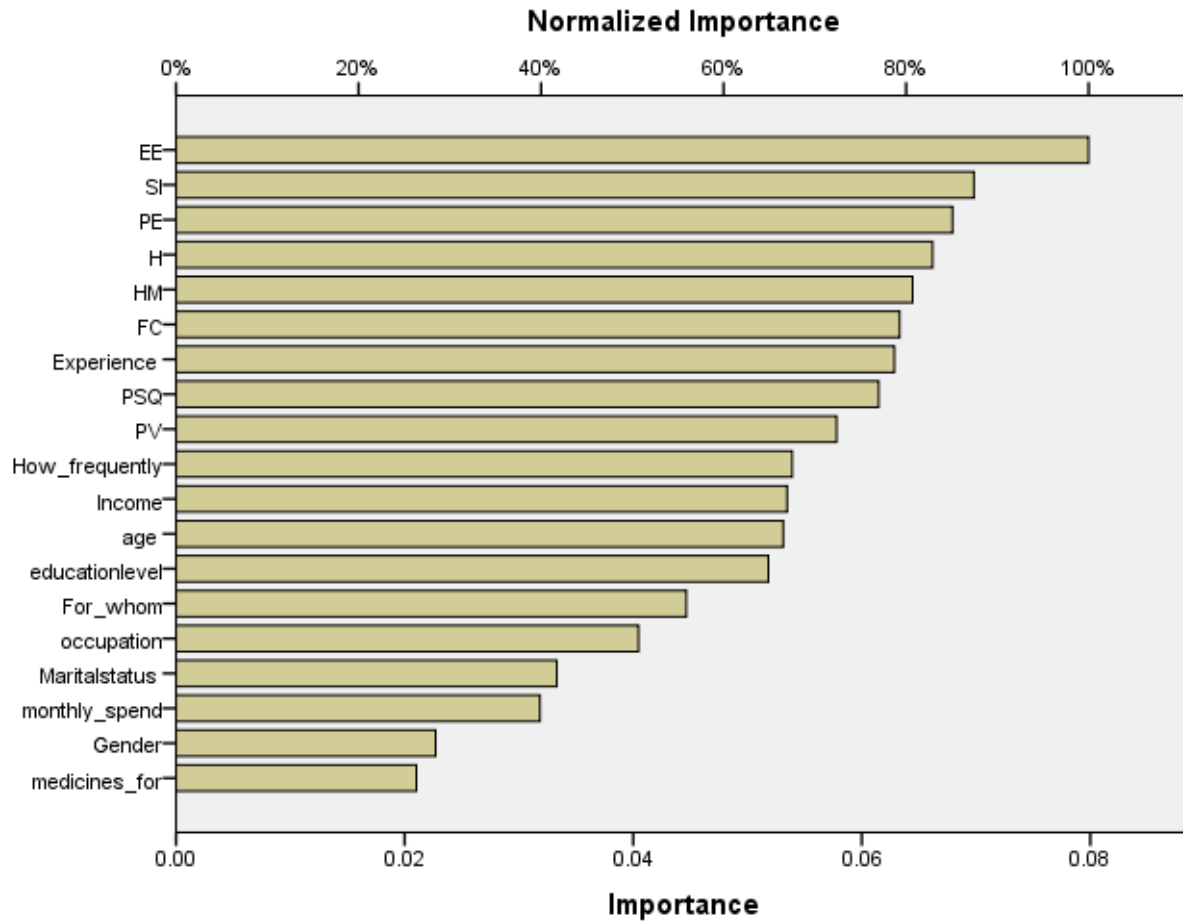


Fig 17. Importance of variables using ANN

The fig17 shows a diagrammatic representation of importance of variables with respect to adoption intention of online pharmacy. People who guess they have more benefits on online pharmacy.

6.2.2 willingness of adoptions intention of online pharmacy

Here in logistic regression analysis, which is a statistical method used to examine the relationship between a set of predictor variables and an outcome variable that has only two possible outcomes (e.g., yes/no)

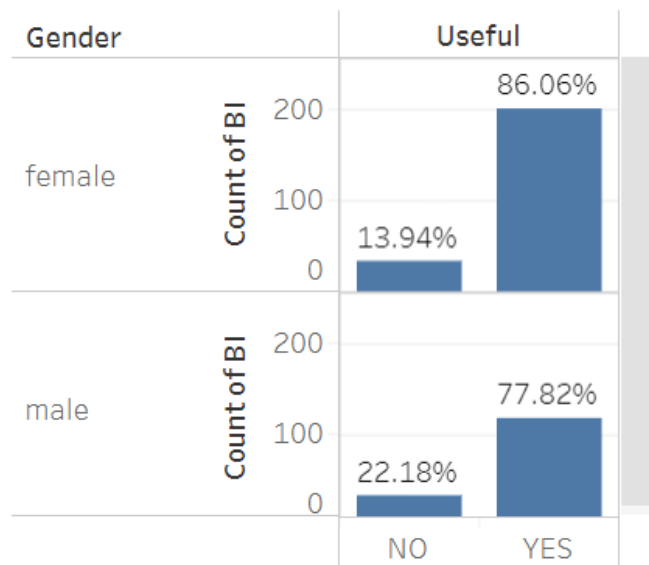


Fig 18. Intention to adopt online pharmacies with respect to gender.

Observed	Predicted		
	NO	YES	Percentage Correct
NO	15	49	23.4%
YES	12	304	96.2%

Overall Percentage	0.0%	100.0%	83.9%
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Table 4. Accuracy for Logistic regression for adoption intention of online pharmacy

	B	S. E	Wald	df	Sig.	Exp(B)
Gender	-.595	.301	3.904	1	.048	.552
Educational level	1.236	.682	3.285	1	.030	3.441
Frequency of medicine	4.558	.568	1.219	1	.027	1.872
Medicines purchased for whom	-3.874	1.395	7.714	1	.005	.633
Performance Expectancy	-.579	.217	7.125	1	.008	.560
Effort expectancy	-.624	.234	7.124	1	.008	.536
Price value	.372	.182	4.159	1	.041	1.4151

Table 5. Logistic regression for intention to adopt online pharmacy.

From Table 5, the statistical analysis suggests that these are the independent variables that are significant predictors of the dependent variable adoption intention of online pharmacies (at the level of significance .05)

6.2.3 Factors associated with the Adoption intention of online pharmacy.

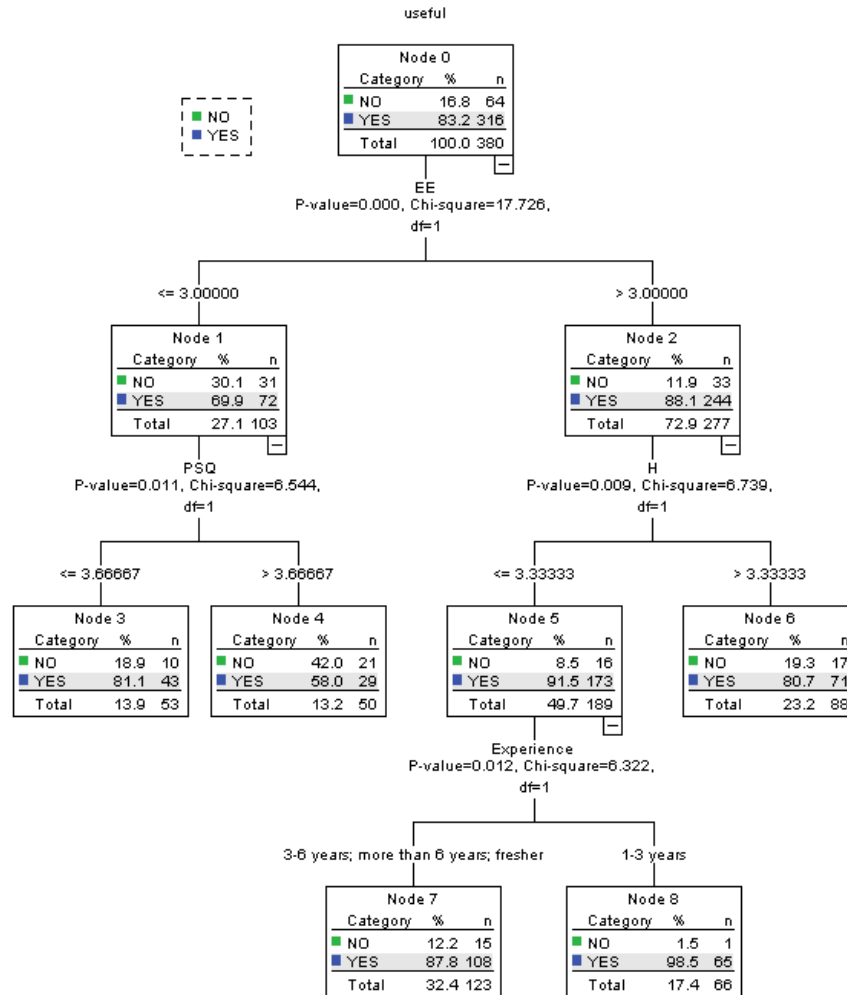


Fig 19. Decision tree for adoption intention

Observed	Predicted		
	NO	YES	Percentage Correct
NO	0	64	0.0%
YES	0	316	100.0%
Overall Percentage	0.0%	100.0%	83.2%

Table 6. Accuracy of Decision tree

Results	Independent Variables Included	EE, PSQ, H, Experience
	Number of Nodes	9
	Number of Terminal Nodes	5
	Depth	3

Table7.Model summary result of Decision tree

The independent variable (Table 7) included in this model were Effort expectancy, Price quality service, habit, and experience, these are most important to predict the adoption intention of online pharmacy.

6.2.4 Intention to buy in online pharmacy.

Our surveillance being conducted. When asked about what is the intention to adopt online pharmacies, to the respondents. They answered many answers. The text was sorted out using sentimental analysis and by using word cloud the answers for the adoption intention of online

pharmacies are online pharmacies are useful and helpful. The most used words found by using word cloud are Easy to buy, using online pharmacies are helpful, good, fast delivery of medicines.

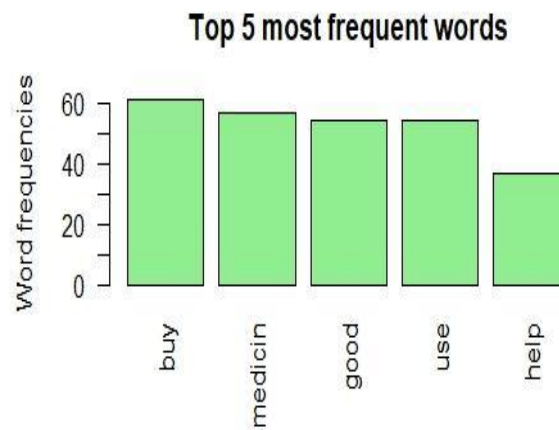


Fig 20. Top 5 most frequent words



Fig 21. Word cloud for adoption intention



Fig 22. Word cloud for adoption intention of online pharmacy

6.2.5 Opinion on online pharmacy.

In the survey, asked about the opinion about online pharmacy respondents have shared many opinions. I have done a sentimental analysis for the opinion on adoption and found that the sentiment trust is there in most of the opinions shared. And using word cloud the most used words by them are joy, good, fear and sadness.

	anger	anticipation	disgust	fear	joy	sadness	surprise	trust	negative	positive
1	5	16	5	7	17	10	5	20	16	35

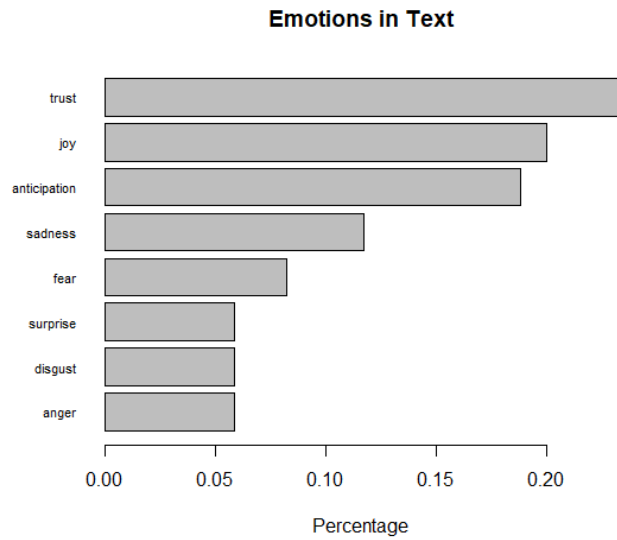


Fig 23. Sentimental analysis for adoption intention of online pharmacy

VII FINDINGS, SUGGESTIONS AND CONCLUSION

7.1 FINDINGS

The main factors like Behavioural intention towards online pharmacies and adoption intention of online pharmacies are studied in this survey from the general population of Coimbatore, we found that 80.6% respondents are having intention to adopt online pharmacy. Logistic regression, Decision tree and Artificial neural networks was done for Adoption intention of Online pharmacies.

According to the findings, several demographic variables such as experience, age, Occupation, income, and education level play a significant role in predicting the adoption intention of online pharmacies.

The UTAUT2 model constructs were found to be important predictors, with effort expectancy, social influence, and performance expectancy being the most influential, followed by habit, facilitating conditions, and hedonic motivation. Price value and perceived service quality were found to be less important in predicting adoption intention.

By considering the accuracy of the algorithms used for the Adoption intention of Online pharmacies like Logistic regression, Artificial Neural Network (ANN) and Decision tree, we see that Logistic regression has an accuracy of 84.4%, Decision tree has an accuracy of 80% and ANN has an accuracy of **85%**. Even though it can be concluded that all the algorithms have high accuracy (i.e., more than 80%), the algorithm ANN has more accuracy compared to the other two.

By using logistic regression, we found that the significant predictors of adoption intention for online pharmacies are PE (Effort Expectancy), EE (Performance Expectancy), PV (Perceived Value), and PSQ (Perceived Service Quality). Specifically, a decrease in PE, EE, and PSQ is associated with an increase in the log odds of adoption intention, while an increase in PV is associated with an increase in the log odds of adoption intention. SI (Social Influence), FC (Facilitating Conditions), HM (Hedonic Motivation), and H (Habit) are not significant predictors in this model, so these variables did not have strong impact on the likelihood of individuals adopting online pharmacies.

So, considering the ANN output (Table 2) we could say that all the variables have importance with respect to the dependent variable. But the three most important variables are considered. They are the Effort Expectancy which has 100% importance, Social Influence which has 87.4% importance, Performance Expectancy which has 85.3% importance So, we could conclude that

people who think that adopting online pharmacies has more benefits and the respondents who consider Effort expectancy as more important are more likely to adopt online pharmacies.

About the opinion on the adoption intention of online pharmacies, respondents have typed down their opinions. With the help of sentimental analysis using R programming, I filtered out the sentiment of the responses and using word cloud we also found the most mentioned word on their opinions. In the analysis found that the Medicines and Useful are the sentiments which are higher in their opinion. Which means that most of the people prefer to buy medicines online and they found these online pharmacies are useful. By using the word cloud, we found out the most used words are buy, medicine, help, good with which we could say that people think that the online pharmacy is good, effective, safe and it could deliver the medicines in doorstep.

As Open-ended question was asked in the survey, in the survey people have asked "what you think about online pharmacies". The same sentimental analysis was done, and the output was derived. The sentiment found was Trust which we could conclude that the people are trusting the online pharmacies and started adopting to the technological changes. With the word cloud we found that the most given words are happy, avail, discounts, trust, buy and safety in which we could conclude that some said that respondents are happy to order medicines.

7.2 SUGGESTIONS

- Online pharmacies should focus on making their websites easy to use and provide accurate information about the products they sell. This will help to increase consumers' confidence in the online pharmacy and make it easier for them to navigate and find what customers need.

- Online pharmacies can increase their perceived value by offering competitive prices, discounts, and loyalty programs. This will make it more attractive for consumers to purchase from the online pharmacy.
- Online pharmacies can improve their perceived service quality by providing excellent customer service, timely delivery, and a user-friendly website.
- Social influence was found to be a significant predictor of adoption intention. Online pharmacies can reach out to influencers such as healthcare professionals, bloggers, and social media personalities to promote their services and build trust with potential customers.
- The research found that demographic variables such as experience, age, income, and education level play a significant role in predicting adoption intention. Online pharmacies should consider these variables when developing marketing strategies and tailor their services to meet the needs of different segments of the population.

7.3 Conclusion

In this research we could conclude that study reveals that online pharmacies can be a popular choice among individuals with higher income and education levels, particularly the younger population. It is suggested that promoting online pharmacy adoption should primarily focus on raising awareness about the importance through influencing by social media. Additionally, implementing strategies that promote habit formation related to using online pharmacies may

prove beneficial. Improving the Price value and service quality of online pharmacies can further enhance the adoption rates.

These findings can serve as a useful guideline for healthcare providers and policymakers in devising effective strategies to encourage the adoption of online pharmacies.

VIII BIBLIOGRAPHY

8.1 APPENDICES/ANNEXURE

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